

Math709 – Foundations of Computational Mathematics II

Spring 2013

Final Exam

You must show all of your work to get a credit for a correct answer.

1. (15 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
 - (a) Find a full SVD for $\mathbf{A}$.
 - (b) Compute $\|\mathbf{A}\|_1$, $\|\mathbf{A}\|_2$, $\|\mathbf{A}\|_\infty$, $\|\mathbf{A}\|_F$.
2. (10 points) Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$. Find the orthogonal projector $\mathbf{P}$ onto $\text{range}(\mathbf{A})$.
3. (10 points) Sketch the proof that every Hermitian, positive definite matrix $\mathbf{A}$ (i.e., $\mathbf{x}^* \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$) has a *unique* Cholesky factorization.
4. (15 points)
 - (a) Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be tridiagonal and hermitian, with all its sub- and superdiagonal entries nonzero. Prove that the eigenvalues of $\mathbf{A}$ are distinct.
 - (b) On the other hand, let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be upper-Hessenberg, with all its subdiagonal entries nonzero. Give an example that shows that eigenvalues of $\mathbf{A}$ are not necessarily distinct.
5. (15 points) Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be a upper Hessenburg matrix and $\mathbf{A} = \mathbf{Q}\mathbf{R}$ be an QR factorization of $\mathbf{A}$.
 - a. What are the structures of $\mathbf{Q}$ and $\mathbf{R}$?
 - b. Show that $\mathbf{RQ}$ is still a upper Hessenburg matrix. (*The preliminary reduction to upper Hessenburg form would be of little use if the steps of the QR algorithm for solving the eigensystem problem does not preserve this structure.*)
6. (10 points) Let $\mathbf{A}, \Delta \mathbf{A} \in \mathbb{R}^{m \times m}$ be symmetric matrices. If $\Delta \mathbf{A}$ is positive definite, show that
$$\lambda_j(\mathbf{A}) \leq \lambda_j(\mathbf{A} + \Delta \mathbf{A}), \quad j = 1, \dots, m$$
where
$$\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_m(\mathbf{A}),$$
$$\lambda_1(\mathbf{A} + \Delta \mathbf{A}) \geq \lambda_2(\mathbf{A} + \Delta \mathbf{A}) \geq \dots \geq \lambda_m(\mathbf{A} + \Delta \mathbf{A}).$$
7. (15 points) Consider a matrix of form $\mathbf{A} = \mathbf{I} + \alpha \mathbf{B} \in \mathbb{R}^{m \times m}$ where $\mathbf{B}$ is real skew symmetric (i.e., $\mathbf{B} = -\mathbf{B}^T$) and α is a nonzero real number.

- a. Show that $(\mathbf{Ax}, \mathbf{x})/(\mathbf{x}, \mathbf{x}) = 1$ for all nonzero $\mathbf{x}$.
- b. Show that the resulting Hessenberg matrix by the Arnoldi iteration will have the following tridiagonal form:

$$\mathbf{H}_k = \begin{pmatrix} 1 & -\eta_2 & & & \\ \eta_2 & 1 & -\eta_3 & & \\ & & \ddots & & \\ & & \eta_{k-1} & 1 & -\eta_k \\ & & & \eta_k & 1 \\ & & & & \eta_{k+1} \end{pmatrix}.$$

(Hint: $h_{ij} = \mathbf{q}_i^T \mathbf{A} \mathbf{q}_j$)

- c. Write out a pseudo code for the simplified Arnoldi iteration applied to $\mathbf{A}$.

8. (10 points)

$$\mathbf{A} = \begin{pmatrix} 0 & & & 1 \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{pmatrix} \in \mathbb{R}^{m \times m}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^m, \quad \mathbf{x}_* = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^m.$$

Clearly $\mathbf{x}_*$ is the exact solution of the linear system $\mathbf{Ax} = \mathbf{b}$. Show how the GMRES method will converge on this linear system. (Hint: check the Krylov subspaces $\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_m$.)